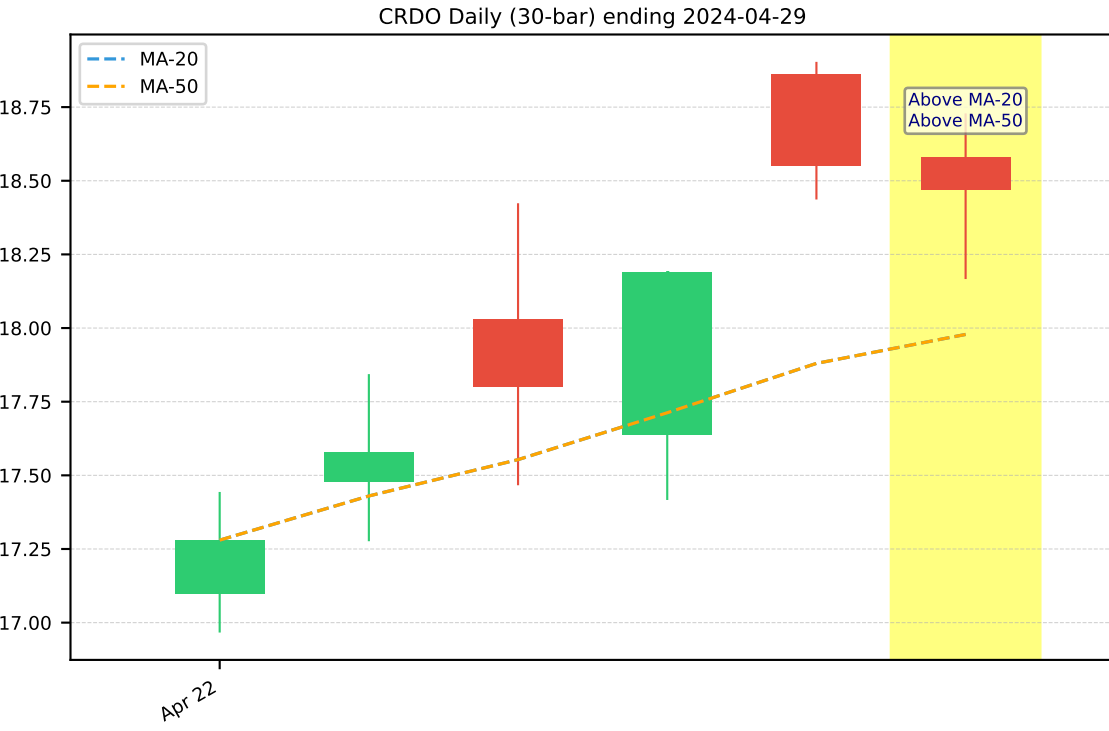
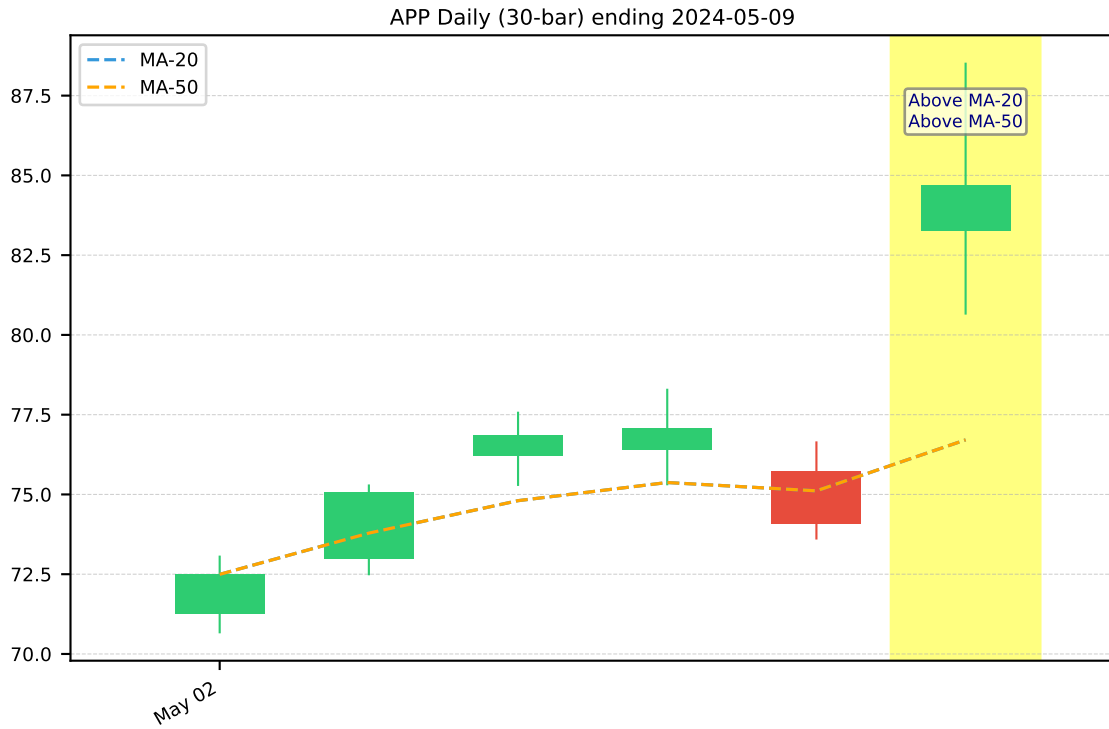
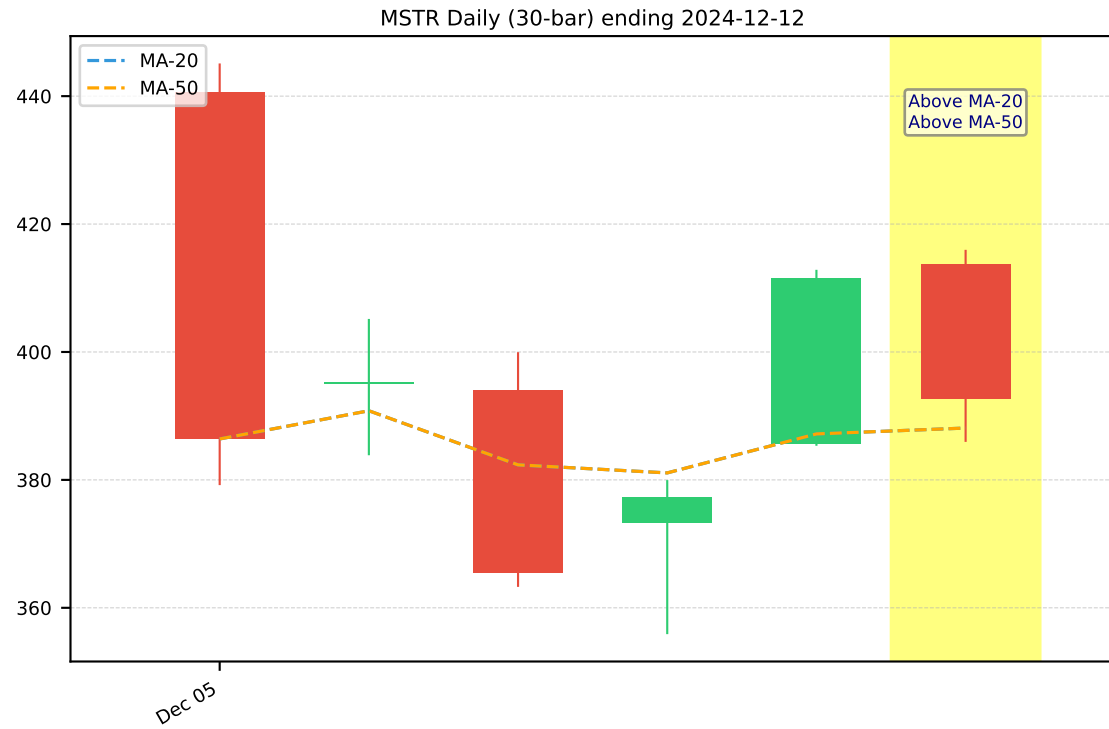


2024 Noise Day Patterns — max single-trade gain < 0.5%
All noise days share: tight first bar, fast fallback exits, direction flips, counter-trend signals



2024 Noise Day Avoidance Rules — Validated Across All 4 Replay Years

Rule 1 — M1 First-Bar Range Gate

Skip M1 entry if first 5-min bar (09:30–09:35) range < 1.5% of bar midpoint.
Noise days averaged 2.37% first-bar range vs 3.22% on good days.

Rule 2 — Direction Flip Block

Once a ticker exits both long AND short on the same day, block further same-ticker entries.
Direction flips cost -37to-272 per additional trade on noise days.

Rule 3 — High-Noise-Ticker Fallback Guard

If a ticker from the list (CVNA, MSTR, COIN, CRDO, APP, AMD, CLS, MRVL) exits via fallback_20pct within 5 min, block all same-ticker entries for the rest of the window.
These tickers appear on noise days in all 4 replay years.

Rule 4 — M1 Volume Gate

Require first 5-min bar volume $\geq 1.5 \times$ (20-day ADV / 78 bars) before any M1 entry.
81% of fallback_20pct exits cluster at 09:50 — weak open = no institutional participation.

Rule 5 — Daily Loss Circuit Breaker

Block new entries once daily realized P&L reaches -\$150. Existing positions unaffected.
Caps worst noise days: 2024 noise days ranged -17to-449 cap P&L.

Full-year 2024 noise metrics: 2024 had 33/252 (13.1%) noise days | Win rate 20-24% vs 41-44% on good days
Fast exits ≤ 5 min: 31-35% | 0f fast exits, losses: 80-83% | M1 fallback rate: 45-49%